

Dr. Habila Basumatary, Ph.D.

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Employment History

- July 2024 – Till date **Assistant Professor, Grade II (Level 10)**, Computer Science and Engineering Department, Indian Institute of Information Technology, Pune, Maharashtra, India.
- Dec 2022 – July 2024 **Post-Doc**, Computer Science and Engineering Department, Indian Institute of Technology (IIT), Guwahati, India.
Thesis title: *AI and IoT based Attack Detection and Authentication Scheme for Cyber Security in Grid Connected Power Electronics Converters*
- Sept 2021 – Nov 2022 **Assistant Professor (Guest)** Computer Science and Engineering Department, School of Engineering, Tezpur University, Assam, India.

Education

- Feb 2018 – Aug 2021 **Ph.D., National Institute of Technology (NIT), Agartala**
Thesis title: *Method to Maximize the Lifetime of Wireless Sensor Network using Single Sink*
- June 2015 – June 2017 **M.Tech., Information Technology, North Eastern Regional Institute of Science and Technology (NERIST)**
- June 2011 – June 2015 **B.Tech., Information Technology, Assam University, Silchar.**
- 2009 – 2011 **XII (Science), K. K. Handique Jr. College, Pathsala**
- 2009 **X, Bijni Bandhab H. S. School, Bijni**

Research Publications

Journal Articles

- 1 H. Basumatary, A. Debnath, M. K. Debbarma, and B. K. Bhattacharyya, "Performance analysis of centroid based routing protocol in wireless sensor networks," *IETE Technical Review*, 2024. [DOI: 10.1080/02564602.2023.2230947](#).
- 2 A. Debnath, H. Basumatary, M. Dhar, M. K. Debbarma, and B. K. Bhattacharyya, "A routing technique for enhancing the quality of service in vanet," *IETE Journal of Research*, vol. 69, 2023. [DOI: 10.1080/03772063.2021.1886879](#).

- 3 H. Basumatary and M. K. Debbarma, "Comparative study on different parameters used for energy conservation in wireless sensor networks," *IETE Technical Review*, vol. 39, 2022.  DOI: 10.1080/02564602.2021.1882344.
- 4 H. Basumatary, A. Debnath, M. K. Debbarma, and B. K. Bhattacharyya, "Centroid-based routing protocol with moving sink node for uniform and non-uniform distribution of wireless sensor nodes," *Journal of Supercomputing*, vol. 77, pp. 3727–3751, 2021.  URL: <https://doi.org/10.1007/s11227-020-03414-8>.
- 5 H. Basumatary, A. Debnath, M. K. Debbarma, and B. K. Bhattacharyya, "Clustering based two-dimensional motion of sink node in wireless sensor networks," *Wireless Personal Communication*, vol. 118, pp. 161–183, 2021.  URL: <https://doi.org/10.1007/s11277-020-08007-5>.
- 6 A. Debnath, H. Basumatary, M. Dhar, M. K. Debbarma, and B. K. Bhattacharyya, "Fuzzy logic-based vanet routing method to increase the qos by considering the dynamic nature of vehicles," *Computing*, vol. 103, pp. 1391–1415, 2021.  URL: <https://doi.org/10.1007/s00607-020-00890-x>.
- 7 H. Basumatary and M. K. Debbarma, "Analysis of mobile sink based routing protocols in wireless sensor networks," *International Journal of Computational Intelligence and IoT*, vol. 2, no. 3, 2019.  URL: <https://ssrn.com/abstract=3358300>.
- 8 A. Debnath, H. Basumatary, T. Anirban, M. K. Debbarma, and B. K. Bhattacharyya, "Center of mass and junction based data routing method to increase the qos in vanet," *AEU - International Journal of Electronics and Communications*, vol. 108, pp. 36–44, 2019.  URL: <https://doi.org/10.1016/j.aeue.2019.05.026>.
- 9 M. M. Singh and H. Basumatary, "MERAM-R: Multi-clustered energy efficient routing algorithm with randomly moving sink node," *Journal of Scientific and Industrial Research*, pp. 15–17, 2018.  URL: <http://nopr.niscpr.res.in/handle/123456789/43315>.

Conference Proceedings

- 1 H. Basumatary, R. Chowdhuri, and M. K. Debbarma, "A review on recent techniques developed for energy conservation in wireless sensor networks," in *Proceedings of the International Conference on Intelligent Systems, Advanced Computing and Communication (ISACC-2023)*, Silchar, India, 2023, pp. 1–6.  DOI: 10.1109/ISACC56298.2023.10083645.
- 2 H. Basumatary and M. M. Singh, "Multi clustered energy efficient routing algorithm with mobile sink node moving in clockwise direction," in *Proceedings of the International Conference on Recent Advancement in Computer and Communication (ICRAC-2017)*, vol. 34, Bhopal, India, 2018, pp. 645–654.  URL: https://link.springer.com/chapter/10.1007/978-981-10-8198-9_67.

Skills

Languages	 English, Hindi, Boro, Assamese, Bengali.
Coding	 C, C++, Java, Python, SQL, ALP, MATLAB, $\text{\LaTeX}$
Web Dev	 HTML, CSS, JavaScript

Internship Experience

- 2014  2 Months (June - July, 2014) Internship on Advanced Java Platform at GenX Soft Technologies Pvt. Ltd., Jaipur Rajasthan India.

Workshop and Training Experience

- 2019  One-week AICTE Training and Learning (ATAL) Academy Workshop on “Artificial Intelligence” at NIT Agartala from 22-02-2019 to 26-02-2019.
- 2018  One-week National Workshop on Soft Computing and its Application in Real Life Problems organized by Department of Mathematics, NIT, Agartala from 09-04-2018 to 13-04-2018.
- 2016  3rd IEEE Workshop on Advanced MATLAB Applications - “ADMAT-2016” organized by IIT Guwahati from 01-10-2016 to 02-10-2016.
 Workshop on Cognitive Radio Networks: Prospects and Challenges organized by Department of CSE, NERIST under ITRA Project (“Mobile Broadband Service Support over Cognitive Radio Networks” in collaboration with IIT, Delhi, LMNIT Jaipur and IRPE, Kolkata) on 8th April, 2016.
- 2013  Workshop on Adrodizer (Android Application Making Workshop) and PHP PRO (PHP MYSQL Web Development Workshop) organized by Department of Information Technology in collaboration with IPEG Solutions from 26th to 28th December, 2013.
- 2012  Workshop on Android Application Development held at Assam University, Silchar, India, on 10-11 March, 2012.
- 2011  HackIN workshop on Ethical Hacking and Information Security organized by Horizon (A unit of MBS Group) held at Assam University, Silchar from 24th to 25th September, 2011.

Miscellaneous Experience

Awards and Achievements

- 2009  Awarded with Anandaram Boruah Scheme Award for securing first division in the 10th standard board examination by State Government of Assam.

Extra-Curricular Activities

- 2010  NCC ‘B’ certificate by unit 62 Assam (G) Bn, NCC with No CSW/11/12267 with Rank: Sgt

Professional Responsibilities

Reviewer

- 2023  International Conference on Intelligence Systems, Advanced Computing and Communications (ISACC), held at Assam University Silchar, India from 03-04 February, 2023.
 4th International Conference on Frontiers in Computing and Systems (COMSYS 2023), held at IIT Mandi, Himachal Pradesh, India from 16-17 October, 2023.

Membership

-  Associate Member of the Institution of Engineers (IEI), India.
-  Student Member OF Institute of Electrical and Electronics Engineers (IEEE)